Is Neural Voice Conversion a Robust Speaker Anonymizer?

Classical McAdams Warping versus Zero-Shot Neural Voice Conversion under Different Speaker-Verification Attackers

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Code: `https://github.com/Max9i/elec5305-project-550145574`

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Abstract

Speaker anonymization is usually judged by the equal error rate (EER) of a single speaker-verification attacker, so the privacy claim depends on that attacker. This report tests how robust such claims are. I compare a classical McAdams anonymizer, implemented in MATLAB with LPC pole warping, with the pretrained zero-shot voice conversion model Seed-VC in its anonymization-only mode. Both are evaluated on LibriSpeech test-clean (40 speakers, 8000 trials) against two attackers, ECAPA-TDNN and WavLM-SV, each at three levels of knowledge about the anonymizer. Utility is the word error rate (WER) of wav2vec 2.0. The ranking of the two systems never changes: Seed-VC gives EERs of 39–48 % at a WER cost of 0.6–0.7 points, far more privacy than any McAdams setting. The amount of privacy, however, depends on the attacker. McAdams looks much more private to WavLM-SV than to ECAPA-TDNN, and most of its privacy disappears when the attacker anonymizes its own enrollment speech (EER drops from 8.0 % to 1.8 % at $\alpha = 0.8$). Seed-VC's weak spot moves to WavLM-SV and to speakers whose pitch is close to its output voice.

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1 Introduction

Speech is recorded and stored everywhere, from voice assistants to call centers and the large datasets used to train speech recognizers. A recording carries the words, but the voice also identifies the speaker, and modern speaker-verification systems can recognize a person from a few seconds of audio. Speaker anonymization changes the voice so that the speaker cannot be recognized, while the words stay usable [1].

The privacy of an anonymizer is usually measured by attacking it with an automatic speaker verification (ASV) system and reporting the equal error rate (EER). This makes privacy a property of the anonymizer and of the attacker. An attacker that knows how the speech was anonymized can be much stronger than one that does not [2], and recent challenge results show how quickly attackers improve [3]. A system that looks private against one attacker may not be private against another.

In this project I compare two very different anonymizers. The first is the classical McAdams method [4], which shifts the formants by warping LPC poles. I implemented it from scratch in MATLAB. The second is Seed-VC [5], a pretrained zero-shot voice conversion model with an anonymization-only mode, which I used without training. Both are reproductions of existing methods. My modification is in the evaluation: instead of one attacker, I use two attacker models of different design, each at three levels of knowledge, and I give speaker-bootstrap confidence intervals for the results. The research question is:

> *How robust is the privacy–utility trade-off of classical and neural voice anonymization when the speaker-verification attacker changes?*

The main contributions are:

- a comparison of McAdams and Seed-VC on one fixed LibriSpeech protocol.
- result that Seed-VC is more robust to changes in the attacker model than McAdams.
- an analysis of which speakers remain identifiable and why, highlighting the role of pitch in Seed-VC's remaining privacy risks.

2 Literature Review

2.1 Speaker anonymization and its evaluation

The VoicePrivacy Initiative made speaker anonymization a shared task with common data, baselines and metrics [1, 6, 7]. Privacy is measured with an ASV attacker, where a higher EER means better privacy, and utility with ASR word error rate (WER).

Privacy also depends on what the attacker knows. Srivastava et al. [2] found that voice conversion may protect against attackers who know little, but not against attackers who know how the conversion was applied. VoicePrivacy uses three levels: an *ignorant* attacker enrolls with original speech, a *lazy-informed* attacker anonymizes its enrollment speech with the same system, and a *semi-informed* attacker also retrains its ASV model on anonymized speech [7].

2.2 Anonymization methods

Signal-processing methods change the voice with a fixed rule. McAdams warping [4] uses linear prediction (LPC) [8] to split speech into an excitation and an all-pole vocal-tract filter, and moves the formants by raising the pole angles to a power α. It needs no training, but a fixed α is one deterministic transform. Patino et al. noted that it gives the same pseudo-voice for enrollment and test speech and is easy to reverse, so they proposed a random α for each utterance. Neural methods learn to replace the voice. The first VoicePrivacy baseline swaps the speaker's x-vector for an average of other speakers' x-vectors and resynthesizes the speech with a neural vocoder [9].

Table 1: Candidate anonymization approaches.

Approach	Strengths	Weaknesses	Used here
McAdams warping [4]	no training, fast, easy to analyze, VoicePrivacy baseline	fixed α is deterministic and reversible; artifacts at strong α	yes, own MATLAB code
x-vector + neural vocoder [9]	strong privacy, VoicePrivacy baseline	needs a trained synthesis pipeline and an x-vector pool	no
Zero-shot VC, Seed-VC [5]	pretrained, open, high quality, anonymization-only mode	not designed or tested for privacy	yes, pretrained

In VoicePrivacy 2022, voice-conversion systems kept utility best, while combining ASR with speech synthesis gave more privacy [6].

Seed-VC [5] is a newer zero-shot voice conversion model: it can imitate any voice from a short sample without speaker-specific training. Its paper describes a diffusion transformer trained with a "timbre shifter" that reduces leakage of the source voice. The V2 model used here is documented only in the code repository [10] and adds an anonymization-only mode. I found no published privacy evaluation of it. Table 1 compares the candidates. I chose McAdams because I could build it from course material and it is a VoicePrivacy baseline, and Seed-VC because it is open, pretrained, runs on one laptop GPU and needs no target speaker.

2.3 Speaker-verification attackers and ASR

ECAPA-TDNN [11] improves on x-vector speaker embeddings with channel attention and features from several layers. It is also the attacker model in VoicePrivacy 2024 [7]. WavLM [12] first learns general speech features from unlabelled audio and is then fine-tuned for verification, so the two attackers look at speech in different ways.

To evaluate speech intelligibility, we use wav2vec 2.0 [13], a self-supervised speech representation model fine-tuned for automatic speech recognition (ASR) on LibriSpeech [14].

3 Methods

3.1 Data and evaluation protocol

All experiments use LibriSpeech test-clean [14]. I used all 40 speakers (20 female, 20 male). For each speaker, a script drew 5 enrollment and 10 trial utterances of 4–15 s with a fixed seed, and the two sets do not overlap. Every trial is scored against every speaker model of the same sex, which gives 400 genuine and 7600 impostor pairs. The lists were saved once, so every anonymizer and every attacker sees exactly the same data. For the main (ignorant) attacker, only the trials are anonymized.

One caveat: enrollment and trial utterances of a speaker come from the same book chapters, so they often share a recording session. This helps the attacker. Section 4.4 checks how much.

3.2 Classical anonymizer: LPC and McAdams warping

I implemented the McAdams anonymizer of Patino et al. [4] in MATLAB. The signal is cut into 20 ms frames with a 10 ms hop, and a square-root Hann window is used for both analysis and synthesis. An order-20 LPC model $A(z)$ is fitted to each frame [8]. The residual $e[n]$, obtained by filtering the frame with $A(z)$, carries the excitation, so it keeps F0 and timing. Each complex pole $re^{j\varphi}$ of $1/A(z)$, with $0 < \varphi < \pi$, is moved to

$$re^{j\varphi^{\alpha}}, \qquad \varphi \text{ in radians}, \tag{1}$$

its conjugate is mirrored, and real poles are kept. The frame is resynthesized by filtering $e[n]$ with the new $1/A'(z)$, and the frames are overlap-added. Finally the output is scaled to the peak level of the input.

The warp keeps the pole radii, so the filter stays stable and the formant bandwidths do not change. The map $\varphi \mapsto \varphi^\alpha$ has a fixed point at $\varphi = 1\,\text{rad}$ ($f_s/2\pi \approx 2.55\,\text{kHz}$). For $\alpha < 1$, F1 and F2 move up toward it, the higher formants move down, and no pole is left above $(f_s/2\pi)\,\pi^\alpha$ (6.4 kHz at $\alpha = 0.8$). Testing showed two side effects. When warped poles crowd together, the filter gain jumps and some frames become loud bursts. A Python port of the VoicePrivacy reference code does the same, so this comes from the method. Also, voicing breaks down at strong α (Section 4.4). I used $\alpha \in \{0.9, 0.8, 0.7, 0.6\}$ with no further tuning. A self-check confirms that $\alpha = 1$ returns the input.

3.3 Neural anonymizer: Seed-VC

Seed-VC [5] is an open zero-shot voice conversion model. I used its V2 model at a fixed commit [10], without any training or fine-tuning.

V2 has an "anonymization-only" mode that converts the input toward an average voice instead of a target speaker. Reading the code showed how this mode works. At each generation step, the decoder combines three predictions with classifier-free guidance (CFG):

$$\hat{v} = v(c, s) + w_i\,[v(c, s) - v(\varnothing)] + w_s\,[v(c, s) - v(c)], \tag{2}$$

where c is the content of the input speech, s is the speaker prompt and style vector, and $v(\cdot)$ is the decoder's prediction given only those conditions. The intelligibility rate w_i pushes the output toward the content, and the similarity rate w_s toward the target speaker. In anonymization-only mode s is zero, so $v(c, s) = v(c)$ and the last term disappears. The similarity rate therefore has no effect, and the only guidance setting left is w_i. I set it to 0.0, 0.7 (the default) and 1.0, and kept everything else fixed: 30 diffusion steps, style conversion off (so timing is kept) and a fixed random seed.

3.4 Attackers and threat models

Two pretrained speaker-verification models act as attackers. The first is ECAPA-TDNN [11]. The second, WavLM-SV, is Microsoft's WavLM Base Plus [12] fine-tuned for verification. The two differ in architecture and training data, so a conclusion shared by both is less likely to depend on one speaker representation. Neither model is retrained. WavLM-SV does not normalize its input level, and the anonymizers change the level, so every waveform is first scaled to the same RMS value. A speaker model is the mean of the unit-length enrollment embeddings, and each trial is scored by cosine similarity. Privacy is the equal error rate (EER) over all 8000 pairs: 0 % means the attacker always succeeds and 50 % means it is guessing.

The attacker can know more or less about the anonymizer [2]. I tested three levels (Figure 1):

- **Ignorant**: original enrollment, anonymized trials. This is the main setting.
- **Lazy-informed**: the attacker knows the system and its setting, and anonymizes its enrollment speech in the same way.
- **Lazy-informed with an adapted backend**: the attacker also runs the same anonymizer, with the same settings, on extra speech of its own (20 utterances from each of the 40 LibriSpeech dev-clean speakers, who are not in the test set). It then trains an LDA on the embeddings of this speech. Enrollment and trial embeddings are projected with the LDA before cosine scoring.

The third level is a cheap stand-in for the semi-informed attacker of VoicePrivacy [7], which retrains the whole verification network on anonymized speech. I did not do this because it means anonymizing a large training set and training a new model for every setting, which needs far more time and GPU resources than I had.

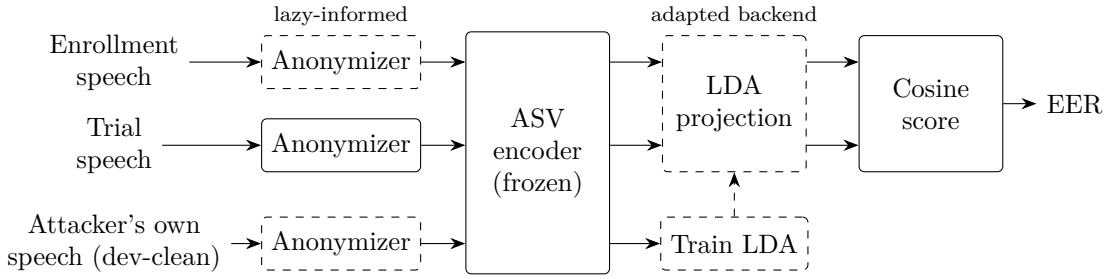


Figure 1: The three attacker levels. Solid blocks are used at every level. The ignorant attacker skips the dashed blocks. The lazy-informed attacker also anonymizes its enrollment speech, and the attacker with an adapted backend also trains an LDA on its own anonymized speech (bottom row).

3.5 Utility and acoustic measures

Utility is the word error rate (WER) of wav2vec 2.0 Large fine-tuned on 960 h of LibriSpeech [13], with greedy decoding and no language model. ΔWER is the WER on anonymized speech minus the WER on original speech. To see what each system changes, I also measured: the median F0 shift in semitones and the correlation of the log-F0 contours (pYIN [15], over frames voiced in both versions). The share of originally voiced frames that are still voiced. And the RMS difference in dB between the gain-normalized order-20 LPC envelopes.

3.6 Statistics

With only 40 speakers, results vary a lot from speaker to speaker. So I give 95 % intervals from a speaker bootstrap [16]: 1000 resamples of 20 female and 20 male speakers, drawn with replacement. The same resamples are used for every condition and attacker, so differences between them are paired.

Raw cosine scores are not comparable across attackers. For the speaker-level analysis, each genuine score s (trial and model from the same speaker) is therefore standardized against the impostor scores (trial and model from different speakers) of the same condition, attacker and sex:

$$z = \frac{s - \mu_{\text{imp}}}{\sigma_{\text{imp}}}, \tag{3}$$

where μ_{imp} and σ_{imp} are the mean and standard deviation of those impostor scores. So z shows how far a genuine trial lies above a typical impostor, and $z \approx 0$ means the attacker cannot tell it apart from one. A trial counts as identified when s is above the 95th percentile of the impostor scores (5 % false acceptance).

3.7 Tools learned

Apart from the LPC work in MATLAB, most tools were new to me. I set up PyTorch on an RTX 5070 Ti, which needed a newer CUDA 12.8 build. I used SpeechBrain and Hugging Face Transformers for the attackers and the ASR, librosa for pYIN, and scikit-learn for the LDA.

4 Results

4.1 Privacy and utility under ignorant attackers

Table 2 and Figure 2 show the main results. On original speech both attackers verify speakers easily (EER 0.25 % and 3.47 %), and the ASR reaches 1.72 % WER. ECAPA-TDNN is clearly the stronger attacker.

McAdams shows a plain trade-off. As α falls, EER rises, but so does WER. $\alpha = 0.9$ changes almost nothing. $\alpha = 0.6$ reaches 29.8 % EER against ECAPA, but WER goes up by 7.6 points.

Table 2: Privacy (EER, ignorant attackers) and utility (WER) on LibriSpeech test-clean. RTF: processing time per second of audio (McAdams on CPU, Seed-VC on GPU).

Condition	ECAPA EER (%)	WavLM-SV EER (%)	WER (%)	ΔWER	RTF
Original	0.25	3.47	1.72	–	–
McAdams $\alpha = 0.9$	0.50	5.25	1.80	+0.08	0.024
McAdams $\alpha = 0.8$	7.97	17.49	2.11	+0.39	0.022
McAdams $\alpha = 0.7$	17.25	31.55	2.93	+1.21	0.022
McAdams $\alpha = 0.6$	29.75	40.25	9.35	+7.63	0.026
Seed-VC CFG 0.0	46.29	47.05	5.13	+3.41	0.260
Seed-VC CFG 0.7	43.75	48.00	2.43	+0.71	0.253
Seed-VC CFG 1.0	44.30	47.53	2.31	+0.59	0.261

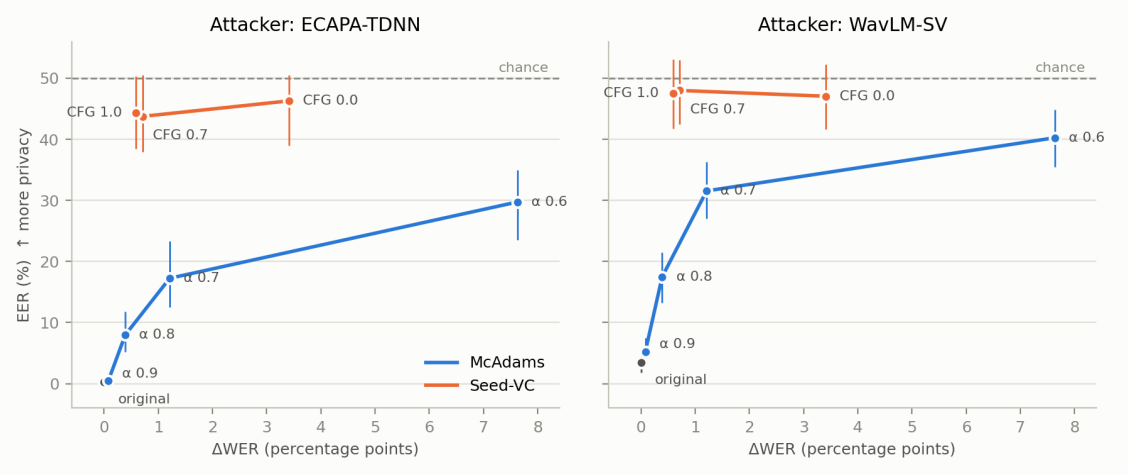


Figure 2: Privacy–utility trade-off against the two ignorant attackers. Error bars: 95 % speaker-bootstrap intervals. Dashed line: chance.

Seed-VC sits in a different place. Both attackers are close to chance (EER 44–48 %), and at CFG 0.7 and 1.0 WER rises by only 0.6–0.7 points. Compared with McAdams at $\alpha \leq 0.7$, Seed-VC is better on both privacy and WER. Only $\alpha = 0.9$ and 0.8 cost less WER, and they give little privacy. CFG 0.0 costs 3.4 points of WER and gives no clear privacy gain, so the intelligibility guidance is worth keeping.

4.2 Agreement between attackers

Figure 3 plots one attacker's EER against the other's. Both attackers rank the conditions in the same order: original, then McAdams from $\alpha = 0.9$ to 0.6, then Seed-VC. The only differences are among the three Seed-VC settings, and they are not stable: only 60–78 % of the bootstrap resamples agree on any of them.

The privacy level is a different story for McAdams. WavLM-SV always reports more privacy than ECAPA. On original speech the gap is 3.2 points (95 % interval 1.8 to 4.5), simply because WavLM-SV is the weaker attacker. Under McAdams the gap grows to 4.8, 9.5, 14.3 and 10.5 points for $\alpha = 0.9$, 0.8, 0.7 and 0.6, and the part above the original gap is positive at every α. In other words, McAdams fools WavLM-SV more than ECAPA. Someone who tested McAdams only with WavLM-SV would report 32 % EER at $\alpha = 0.7$, while ECAPA gives 17 %. For Seed-VC the attackers agree: both intervals reach 50 %, and the gap cannot be told apart from zero.

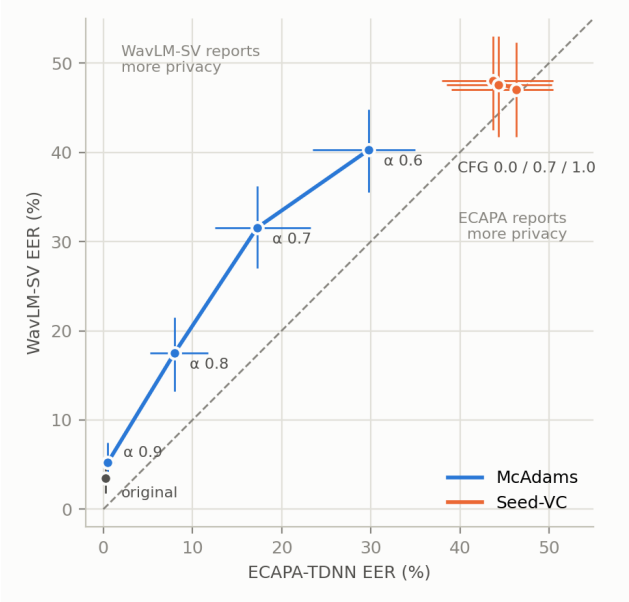


Figure 3: EER of WavLM-SV against EER of ECAPA-TDNN for each condition (ignorant attackers, 95 % intervals). Above the diagonal, WavLM-SV reports more privacy.

Table 3: EER (%) by attacker knowledge: ignorant (Ign.), lazy-informed (Lazy), and lazy-informed with the adapted LDA backend (+LDA). Worst: lowest EER of the six attackers.

	ECAPA-TDNN			WavLM-SV			
Condition	Ign.	Lazy	+LDA	Ign.	Lazy	+LDA	Worst
Original	0.25	–	1.25	3.47	–	4.30	0.25
McAdams $\alpha = 0.9$	0.50	0.75	1.74	5.25	4.27	4.74	0.50
McAdams $\alpha = 0.8$	7.97	1.75	3.53	17.49	7.00	7.31	1.75
McAdams $\alpha = 0.7$	17.25	4.75	7.00	31.55	11.50	11.50	4.75
McAdams $\alpha = 0.6$	29.75	8.69	10.23	40.25	18.01	15.75	8.69
Seed-VC CFG 0.0	46.29	45.80	45.04	47.05	42.71	43.03	42.71
Seed-VC CFG 0.7	43.75	41.50	42.27	48.00	39.25	40.80	39.25
Seed-VC CFG 1.0	44.30	40.50	40.00	47.53	38.97	39.75	38.97

4.3 Informed attackers

Table 3 and Figure 4 add the two informed attackers. Most of McAdams' privacy disappears once the enrollment is anonymized too. At $\alpha = 0.8$, ECAPA's EER falls from 8.0 % to 1.8 %, and even $\alpha = 0.6$ only gives 8.7 %. For $\alpha \leq 0.8$ these drops are significant for both attackers.

Seed-VC keeps most of its privacy. ECAPA changes by less than 4 points, and none of these changes is significant. WavLM-SV gains about 9 points at CFG 0.7 and 1.0 (from 48.0 to 39.3 % and from 47.5 to 39.0 %), and these gains are significant. So WavLM-SV becomes the stronger attacker against Seed-VC, the reverse of the ignorant case.

The adapted LDA backend adds almost nothing. Its only significant gain is for WavLM-SV at $\alpha = 0.6$ (from 18.0 to 15.8 %). For ECAPA it makes things slightly worse, even on original speech (0.25 to 1.25 %), probably because an LDA trained on only 40 speakers throws away some useful dimensions. Taking the worst case over all six attackers, McAdams at $\alpha = 0.8$ keeps only 1.8 % EER, while Seed-VC keeps 39–43 %.

4.4 Sources of residual privacy risk

Acoustic change. Within McAdams, EER rises with the LPC-envelope distance, from 9.3 dB at $\alpha = 0.9$ to 20.3 dB at $\alpha = 0.6$ (Figure 5a). Voicing breaks down at the same time: pYIN finds 72 %

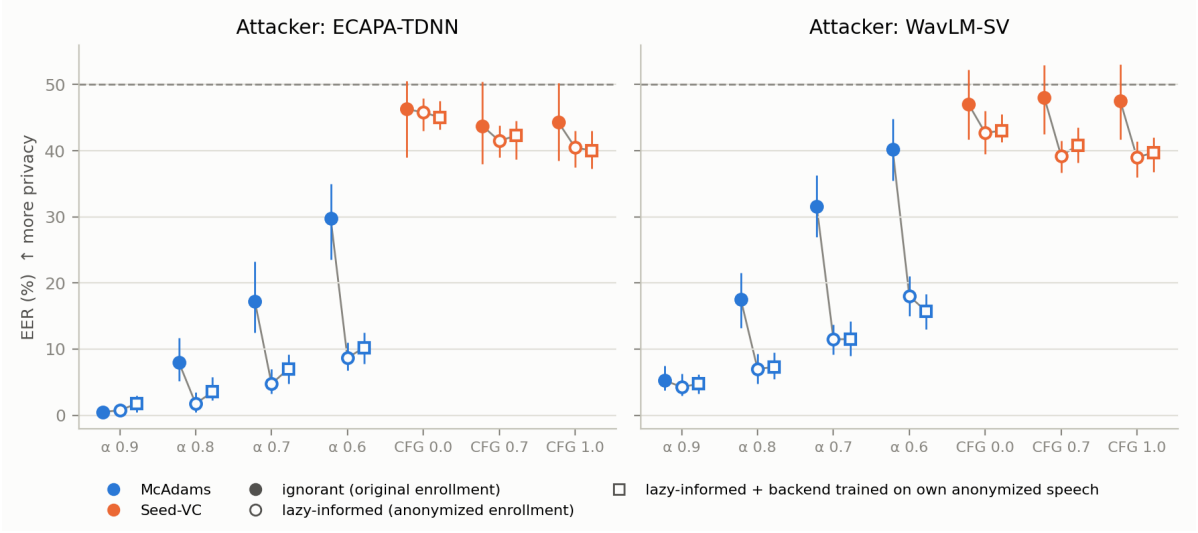


Figure 4: EER at the three knowledge levels. Error bars: 95 % speaker-bootstrap intervals.

of the original voiced frames at $\alpha = 0.9$ but only 7 % at $\alpha = 0.6$. Both measures move with α, so they cannot be separated. Across the two methods, however, envelope change does not explain privacy. Seed-VC changes the envelope *less* than McAdams at $\alpha = 0.9$ (7.8–8.6 dB) but gives far more privacy. The real difference is pitch. McAdams keeps F0 and intonation wherever F0 can still be measured (median shift 0, contour correlation at least 0.97). Seed-VC moves every speaker to about 125 Hz, a median shift of about −3 semitones, and keeps the intonation only weakly (correlation about 0.4).

Figure 6 shows one female trial. Both attackers identify the original and the $\alpha = 0.8$ version. At $\alpha = 0.6$ only ECAPA still does (z-scores 3.4 and 1.0). Seed-VC fools both attackers (z-scores 0.2 and 0.9) with no ASR error, and its spectrogram shows the closer harmonic spacing of a lower voice.

Speaker-level risk. Figure 7 compares the two attackers speaker by speaker. At $\alpha = 0.8$, ECAPA 38 of 40 speakers have a mean z above the threshold and WavLM-SV 18, all of whom ECAPA also identifies. The attackers' speaker rankings agree only weakly (Spearman's $\rho = 0.23$; from 0.48 at $\alpha = 0.9$ down to 0.16 at $\alpha = 0.6$). Each attacker is consistent with itself, though ($\rho = 0.82$ for ECAPA and 0.86 for WavLM-SV between $\alpha = 0.8$ and 0.7). So the two models really do find different speakers hard to hide. Under Seed-VC CFG 0.7, 38 of 40 speakers are identified by neither attacker. The speakers at risk under McAdams $\alpha = 0.8$ are not the ones at risk under Seed-VC CFG 0.7 ($\rho = -0.54$ for ECAPA, 0.05 for WavLM-SV).

Residual risk in Seed-VC. Among female speakers, those with a lower natural pitch, closer to the 125 Hz output, are easier for WavLM-SV to identify ($\rho = -0.43$, Figure 5b). There is no such effect for ECAPA ($\rho = 0.13$) or for male speakers ($\rho = -0.10$).

Session effects. Only 29 trials (10 speakers) come from a chapter not in the enrollment set. For them, the same-chapter advantage is small on original speech and under McAdams, but under Seed-VC it is about as large as the whole remaining signal. Part of what survives Seed-VC may be channel information, though the sample is too small to be sure.

4.5 Computational cost

McAdams runs at a real-time factor (RTF) of 0.022–0.026 on the CPU. Seed-VC needs 0.25–0.26 on the GPU, so it is about ten times slower even with a GPU (Table 2). Both run much faster than real time.

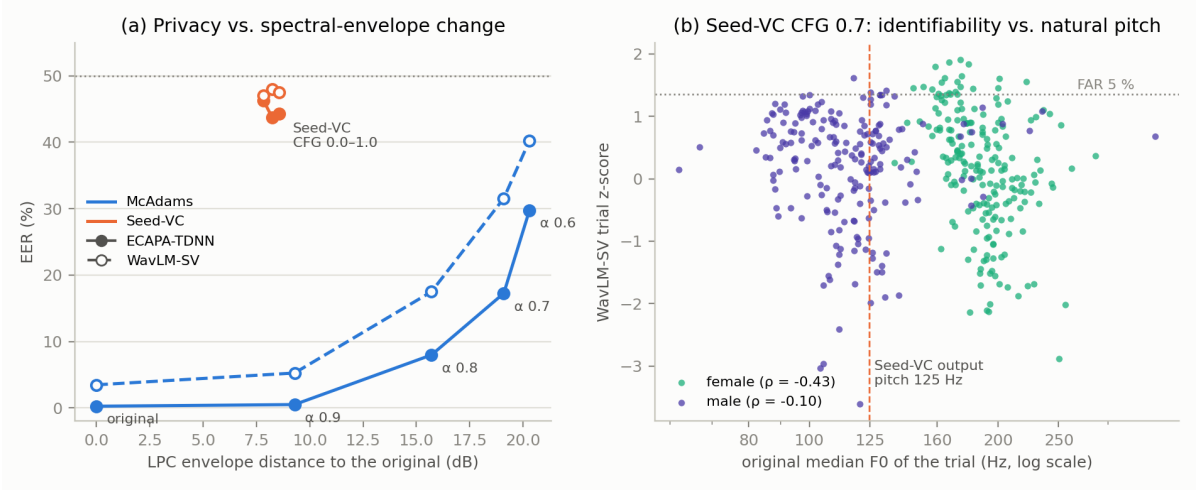


Figure 5: (a) EER against the LPC-envelope distance to the original (filled: ECAPA-TDNN; open: WavLM-SV). (b) Seed-VC CFG 0.7: WavLM-SV z-score of each trial against the original median F0. Dashed line: Seed-VC output pitch.

5 Discussion

5.1 Answer to the research question

Across six attackers (two models at three knowledge levels), the ranking never changed. Seed-VC always gave far more privacy than any McAdams setting, and at CFG 0.7 and 1.0 it cost only 0.6–0.7 points of WER. What did change was the amount of privacy, and which attacker was the most dangerous. For McAdams it was the lazy-informed ECAPA. For Seed-VC it was the lazy-informed WavLM-SV. A claim based on an ignorant ECAPA alone would have overstated McAdams' privacy. A claim based on an ignorant WavLM-SV alone would have overstated it even more, and would also have missed Seed-VC's weak spot.

5.2 McAdams under informed attackers

With a fixed α, McAdams applies the same transform to every speaker. When the attacker passes its enrollment speech through the same transform, enrollment and trials end up in the same warped space, and the differences between speakers survive. So most of the privacy seen by the ignorant attacker came from a mismatch between enrollment and trial, not from removed identity. This is the problem Patino et al. described when they proposed a random α [4], and it agrees with Srivastava et al.'s finding for frequency-warping methods [2].

Published results point the same way, even for the random version. In the VoicePrivacy 2024 evaluation plan, McAdams with α drawn from $[0.5, 0.9]$ gives an average test EER of 4.52 % against the semi-informed attacker, almost the same as 4.59 % for original speech, while WER rises from 1.85 % to 9.95 % [7]. My numbers are not directly comparable: that attacker is retrained on anonymized speech, and its EER on original speech is much higher than mine (0.25 %). But both results say that McAdams gives little privacy once the attacker adapts to it.

5.3 Basis of Seed-VC's privacy

Seed-VC resynthesizes the whole signal and moves every speaker to a similar average voice at about 125 Hz. There is no speaker-preserving transform for an informed attacker to undo in the same way, which is why its EER dropped by at most 9 points. The leftover risk is tied to pitch: female speakers whose natural pitch is close to the output voice were easier for WavLM-SV to identify. So pitch still carries some identity even after conversion. Seed-VC's small WER cost also fits the VoicePrivacy 2022 finding that voice-conversion systems keep utility best [6].

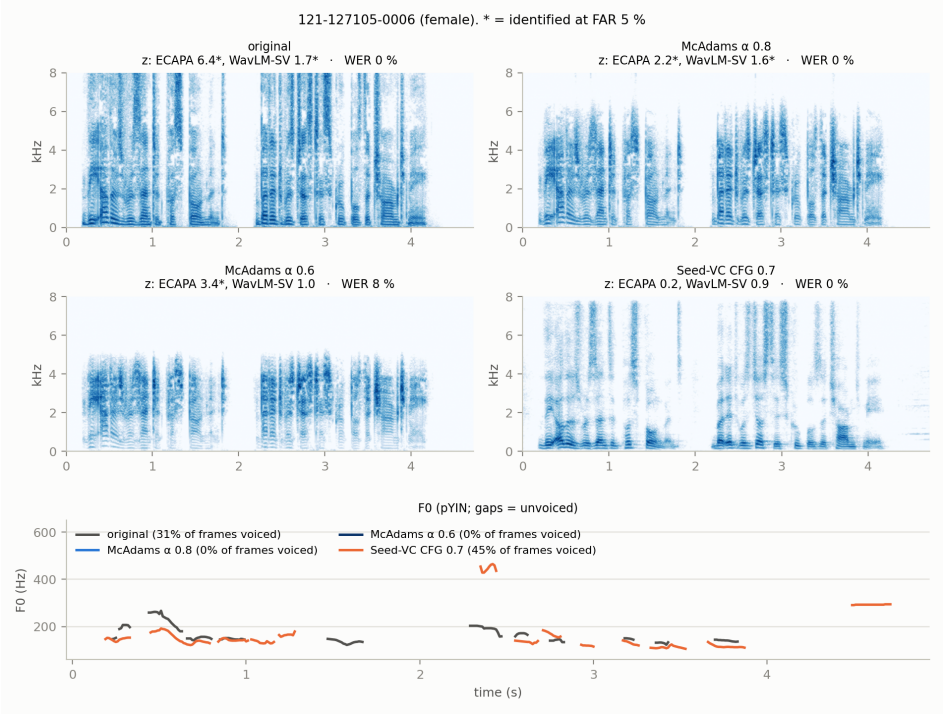


Figure 6: Example female trial (121-127105-0006). Spectrograms with each attacker's z-score (* = identified at 5 % false acceptance) and the WER, and pYIN F0 tracks (gaps: unvoiced).

The main caveat is that all my attackers are frozen. Retrained attackers are much stronger, and in the Attacker Challenge they cut EERs by 25–44 % relative to the baseline attacker [3]. The worst-case EER of about 39 % should therefore be read as an upper bound on Seed-VC's privacy, not as a guarantee. The adapted LDA backend did not lower it further, but a linear backend trained on 40 speakers is a weak test.

5.4 Lessons learned

Fixing the trial lists once and using the same bootstrap resamples for every condition worked well. It made small differences testable, for example the WavLM-SV gain against Seed-VC. Using two attackers from the start was also useful, because the main finding only appears when they are compared.

Several things went wrong at first and changed the analysis:

- Pooling both sexes when computing z-scores inflated the male scores, because the Seed-VC voice sounds male. Per-sex normalization removed this artifact.
- The similarity CFG of Seed-VC has no effect in anonymization-only mode. I only found this by reading the code, which saved a parameter sweep that would have shown nothing.
- By ear, Seed-VC sounded clearest, but its samples were also 6–17 dB louder, so part of "clearer" was "louder". McAdams at $\alpha = 0.6$ was hard to understand by ear, yet the ASR still reached 9.35 % WER. ΔWER from a strong ASR understates how much McAdams damages the speech for human listeners.
- An early conclusion that Seed-VC beats McAdams on both axes had to be narrowed to $\alpha \leq 0.7$.

5.5 Limitations

- No full semi-informed attacker: the neural attackers were never retrained on anonymized speech. I wished to do this, but it would have required anonymizing a large dataset and

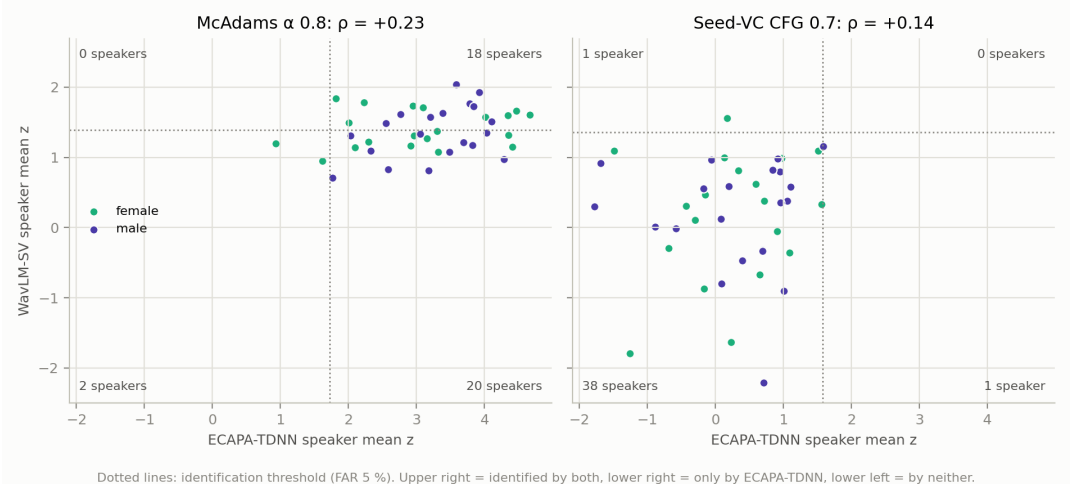


Figure 7: Mean z-score of each speaker, ECAPA-TDNN against WavLM-SV. Dotted lines: identification threshold (5 % false acceptance).

training a separate model for each setup, which was too expensive.

- Only 40 test speakers, so some intervals are wide (for Seed-VC CFG 0.7, ECAPA's EER interval is 38–50 %).
- Enrollment and trial utterances share recording sessions, which helps the attacker.
- EER is the only privacy metric. Other metrics, such as linkability, were not measured.
- Utility is ASR WER only: there is no listening test and no check of emotion, which VoicePrivacy 2024 includes [7]. This is difficult to implement in the report because it may require testers to be trained.
- One corpus of clean read English, and a protocol that differs from VoicePrivacy, so absolute numbers are not comparable with the challenge.
- The processing times come from different hardware.

5.6 Future work

The most useful next step is a real semi-informed attacker: retrain ECAPA-TDNN on speech anonymized by each system and see whether Seed-VC's privacy holds. The same six attackers should also be run on McAdams with a random α, to see how much of the informed-attacker problem it fixes. Further steps are linkability metrics, a small listening test with level-matched samples, an emotion check, and comparing different target voices for Seed-VC.

6 Conclusion

I compared a classical McAdams anonymizer, written from scratch in MATLAB, with the pretrained Seed-VC model on one fixed LibriSpeech protocol, and tested both against two attacker models at three knowledge levels. The ranking held in every case: Seed-VC gave far more privacy at a small WER cost. The amount of privacy did not hold. McAdams looked private to an ignorant attacker, especially WavLM-SV, but most of that privacy disappeared once the attacker anonymized its own enrollment speech. Seed-VC kept an EER of about 39 % or more, but its weak spot moved to WavLM-SV and to speakers whose pitch is close to its output voice. Of the two systems tested, Seed-VC is the better choice, but its privacy has only been tested against frozen attackers. More generally, a privacy claim should be checked against more than one attacker and more than one level of attacker knowledge, and the worst case should be reported.

References

[1] N. Tomashenko, B. M. L. Srivastava, X. Wang, E. Vincent, A. Nautsch, J. Yamagishi, N. Evans, J. Patino, J.-F. Bonastre, P.-G. Noé, and M. Todisco, "Introducing the VoicePrivacy initiative," in *Proc. Interspeech*, 2020, pp. 1693–1697.

[2] B. M. L. Srivastava, N. Vauquier, M. Sahidullah, A. Bellet, M. Tommasi, and E. Vincent, "Evaluating voice conversion-based privacy protection against informed attackers," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 2802–2806.

[3] N. Tomashenko, X. Miao, E. Vincent, and J. Yamagishi, "Privacy attacks on voice anonymization systems: Overview and key findings from the first VoicePrivacy attacker challenge," *Computer Speech & Language*, vol. 102, p. 102036, 2027.

[4] J. Patino, N. Tomashenko, M. Todisco, A. Nautsch, and N. Evans, "Speaker anonymisation using the McAdams coefficient," in *Proc. Interspeech*, 2021, pp. 1099–1103.

[5] S. Liu, "Zero-shot voice conversion with diffusion transformers," arXiv:2411.09943, 2024.

[6] M. Panariello, N. Tomashenko, X. Wang, X. Miao, P. Champion, H. Nourtel, M. Todisco, N. Evans, E. Vincent, and J. Yamagishi, "The VoicePrivacy 2022 challenge: Progress and perspectives in voice anonymisation," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 3477–3491, 2024.

[7] N. Tomashenko, X. Miao, P. Champion, S. Meyer, X. Wang, E. Vincent, M. Panariello, N. Evans, J. Yamagishi, and M. Todisco, "The VoicePrivacy 2024 challenge evaluation plan," arXiv:2404.02677, 2024.

[8] J. Makhoul, "Linear prediction: A tutorial review," *Proceedings of the IEEE*, vol. 63, no. 4, pp. 561–580, 1975.

[9] F. Fang, X. Wang, J. Yamagishi, I. Echizen, M. Todisco, N. Evans, and J.-F. Bonastre, "Speaker anonymization using X-vector and neural waveform models," in *Proc. 10th ISCA Speech Synthesis Workshop (SSW 10)*, 2019, pp. 155–160.

[10] S. Liu, "Seed-VC," https://github.com/Plachtaa/seed-vc, 2025, commit 51383ef (2025-04-20), V2 model.

[11] B. Desplanques, J. Thienpondt, and K. Demuynck, "ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification," in *Proc. Interspeech*, 2020, pp. 3830–3834.

[12] S. Chen, C. Wang, Z. Chen, Y. Wu, S. Liu, Z. Chen, J. Li, N. Kanda, T. Yoshioka, X. Xiao, J. Wu, L. Zhou, S. Ren, Y. Qian, Y. Qian, J. Wu, M. Zeng, X. Yu, and F. Wei, "WavLM: Large-scale self-supervised pre-training for full stack speech processing," *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, 2022.

[13] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020.

[14] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "Librispeech: An ASR corpus based on public domain audio books," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2015, pp. 5206–5210.

[15] M. Mauch and S. Dixon, "PYIN: A fundamental frequency estimator using probabilistic threshold distributions," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2014, pp. 659–663.

[16] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York: Chapman & Hall, 1993.

A Additional figures

All other figures, including the score distributions of every condition, are in the repository under `results/figures/`.

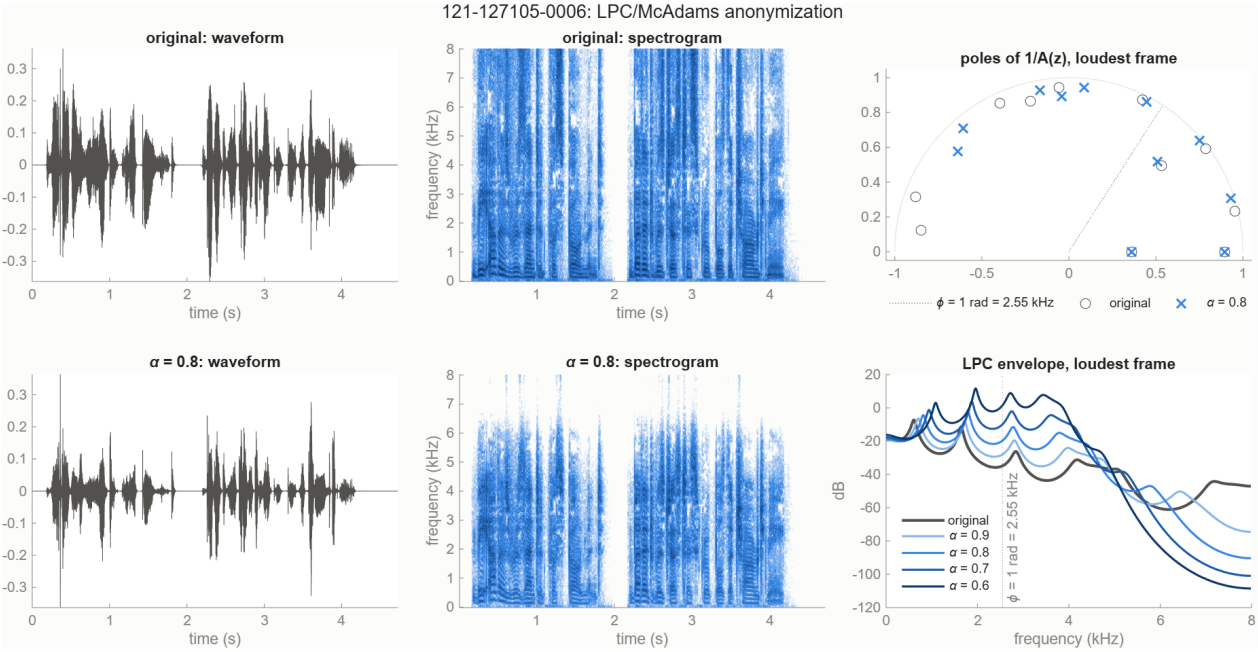
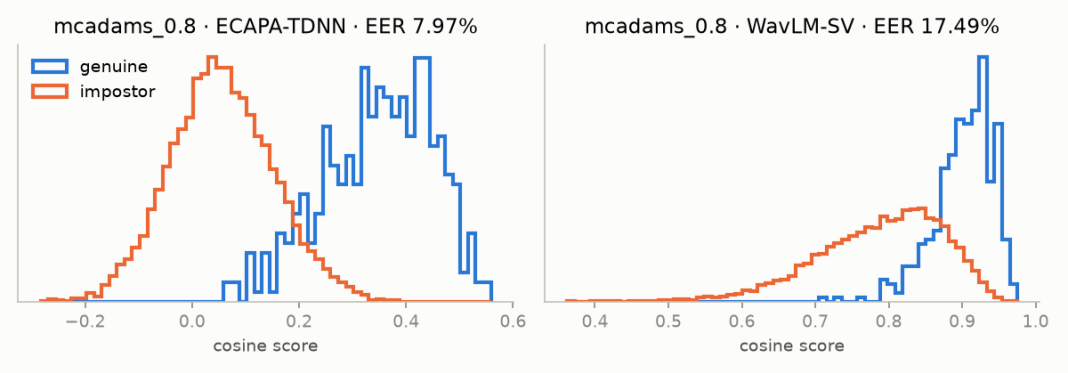
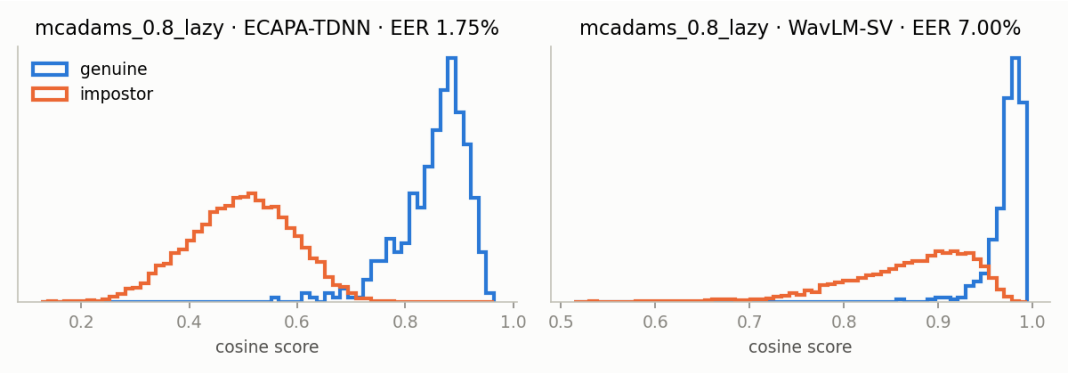


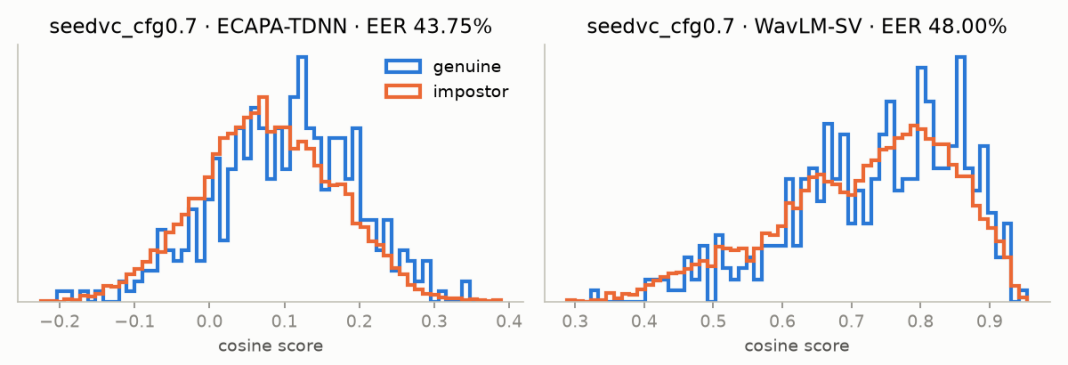
Figure 8: McAdams anonymization of the example utterance in Figure 6. Left and middle: waveform and spectrogram of the original and the $\alpha = 0.8$ version. Right: LPC poles of the loudest frame before and after warping, and the LPC envelope of that frame for every α. Dotted line: the fixed point $\varphi = 1\,\text{rad}$ (2.55 kHz).



(a) McAdams $\alpha = 0.8$, ignorant attackers



(b) McAdams $\alpha = 0.8$, lazy-informed attackers



(c) Seed-VC CFG 0.7, ignorant attackers

Figure 9: Genuine and impostor cosine-score distributions.